SwipeX Milestone 3 — Day 7 Final Review and Milestone 4 Handoff

Domain 4 ownership

Domain 4 owns:

- Resume PDF/DOCX text extraction
- Resume skill extraction
- ATS keyword scoring
- Semantic similarity scoring
- Resume optimization suggestions
- Personalized job recommendation ranking

Completed modules

Resume analyzer

Extracts text from PDF and DOCX files and identifies resume skills using NLP and a curated skills dictionary.

ATS scoring

Compares resume skills and keywords with job required skills and keywords. It produces:

- Keyword-match score
- Semantic similarity score
- Blended ATS score
- Missing skills
- Missing keywords

Resume suggestions

Generates suggestions for missing skills, missing keywords, short resumes, and general improvements. A deterministic rule-based fallback is always available; an LLM is optional.

Recommendation engine

```
Plain Text
```

```
recommendation_score =
    0.70 * semantic_similarity_score
  + 0.30 * swipe_preference_score
```

Swipe preference values:

Action	Score
`like`	1.0
`apply`	1.0
`skip`	0.0
No history or unavailable API	0.5

Scores are normalized to $[0, 1]$.

Domain 2 integration

The recommendation engine retrieves swipe history using the environment variables below:

Plain Text

```
JOB_SERVICE_URL
SWIPE_HISTORY_PATH
```

The agreed contract is:

Plain Text

```
GET /api/v1/swipes/history
X-User-Id: <user_id>
```

Expected response:

JSON

```
[
  {"job_id": 1, "action": "like"},
  {"job_id": 2, "action": "skip"}
]
```

If the Swipe API is unavailable, ranking continues with neutral swipe score `0.5`. The production handoff must confirm Domain 2's actual reachable URL, endpoint path, response fields, and authentication requirements.

Outputs for other domains

Frontend / Intern 1

Suggestions contain:

JSON

```
{
  "type": "missing_skill",
  "priority": "high",
  "item": "Docker",
  "suggestion": "If you have experience with Docker, add it to your Skills sect
}
```

Recommendation results contain:

JSON

```
{
  "job_id": 1,
  "title": "Backend Developer",
  "semantic_similarity_score": 0.825,
  "swipe_preference_score": 1.0,
  "recommendation_score": 0.8775
}
```

Analytics / Milestone 4

Analytics can consume:

- User ID
- Job ID
- ATS score
- Keyword score
- Semantic similarity score
- Recommendation score
- Recommendation rank

- Missing-skill count
- Missing-keyword count
- Swipe action

Possible analytics include average ATS score, frequently missing skills, frequently missing keywords, recommendation-score distribution, click-through rate, and like-to-apply conversion.

Notifications / Milestone 4

Notifications can consume:

- User ID
- Job ID
- Job title
- Recommendation score
- Resume improvement suggestions
- New matching-job events

Example messages:

- `Your resume is missing Docker for this role.`
- `A new job matches your profile with an 87% recommendation score.`
- `Your ATS score may improve if you add relevant REST API experience.`

Known limitations

1. The rule-based suggestion generator is the guaranteed free fallback; an LLM is optional.
2. Suggestions must not claim that a user has a skill without evidence in the resume.
3. Recommendation quality depends on the embedding model and job-text quality.
4. If Domain 2 is unavailable, the system uses neutral swipe score `0.5`.
5. Job IDs in jobs data and swipe history must match.
6. The final Domain 2 URL and response contract must be confirmed before production integration.
7. Semantic scores must be normalized to `[0, 1]`.
8. Production deployment requires authentication, HTTPS, logging, and rate limiting.

Validation

The Day 5 unit suite passed:

```
Plain Text

15 passed
```

Covered behavior includes missing-item suggestions, empty input, fallback behavior, like/apply/skip mapping, neutral history, score blending, and score clamping.

Milestone 4 handoff

Milestone 4 Analytics and Notifications should consume structured service outputs rather than importing internal implementation details. The required integration inputs and outputs are documented above.